Graphing linear inequalities in two variables

1 State whether the given coordinates are solutions to the inequalities:

a $(6, 20)$; $y \geq 4x + 5$

b $(3, 2)$; $3x + 2y \geq 12$

2 State whether each of the given points satisfies the following inequalities:

a $x - y > 7$

i $(-1, -7)$ **ii** $(12, 16)$ **iii** $(13, 6)$ **iv** $(8, -5)$

b $y > 2x$

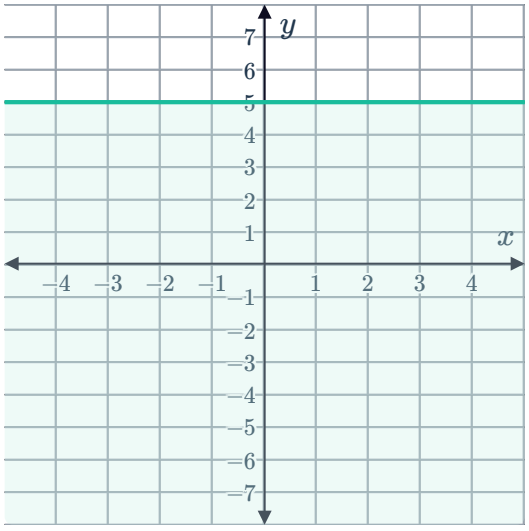
i $(9, 15)$ **ii** $(-1, -2)$ **iii** $(0, 0)$ **iv** $(1, 4)$

c $y > -10x - 2$

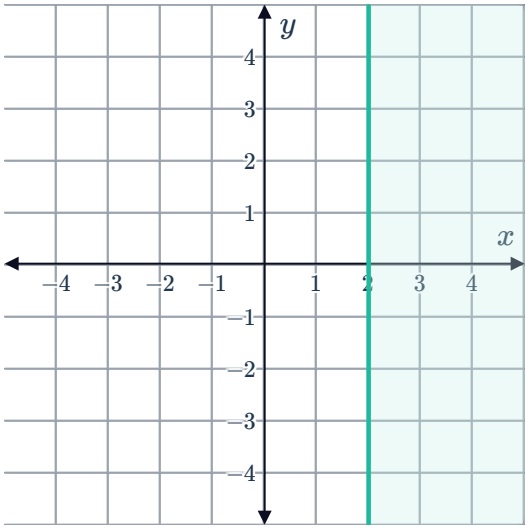
i $(5, -52)$ **ii** $(-1, 3)$ **iii** $(-3, 26)$ **iv** $(1, -11)$

3 Write the inequality that describes the shaded region in the following graphs:

a

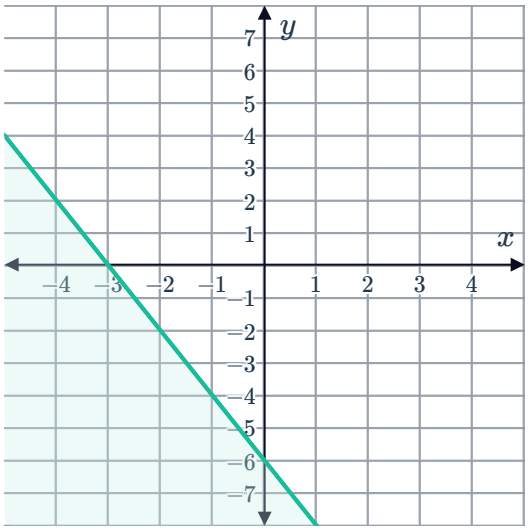


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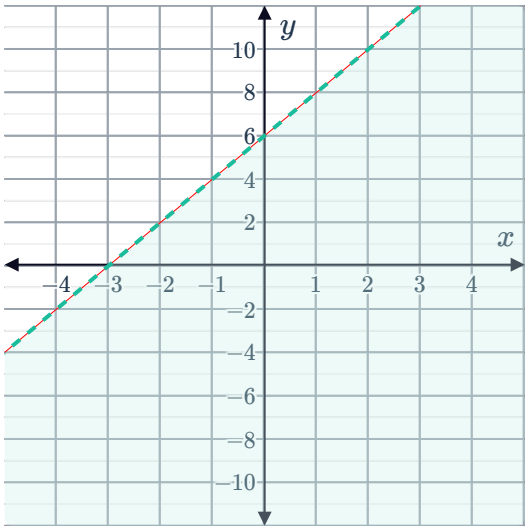


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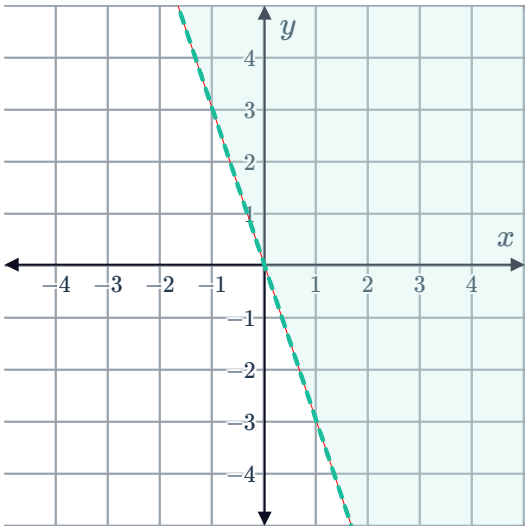
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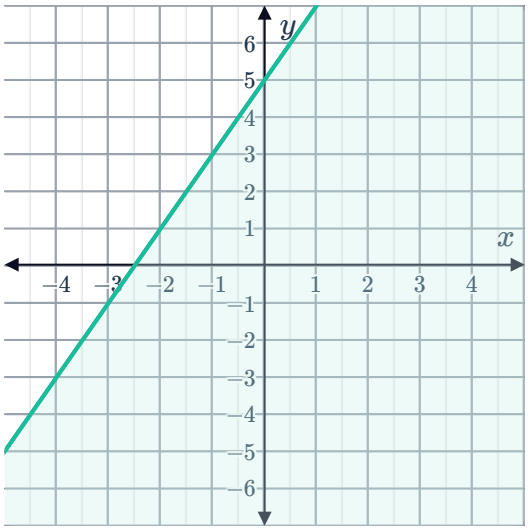
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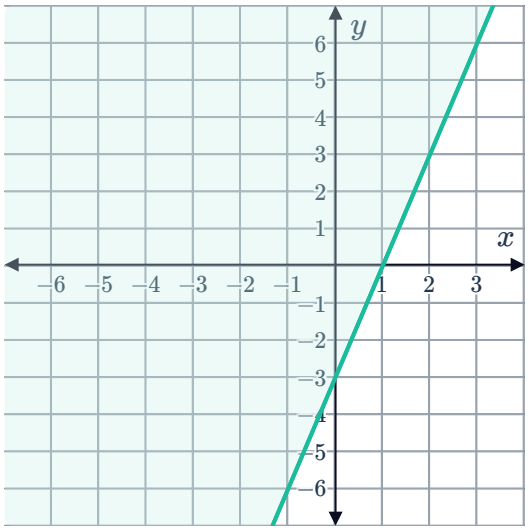
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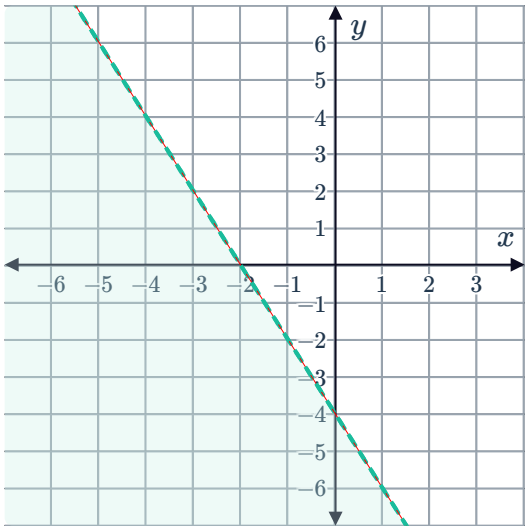
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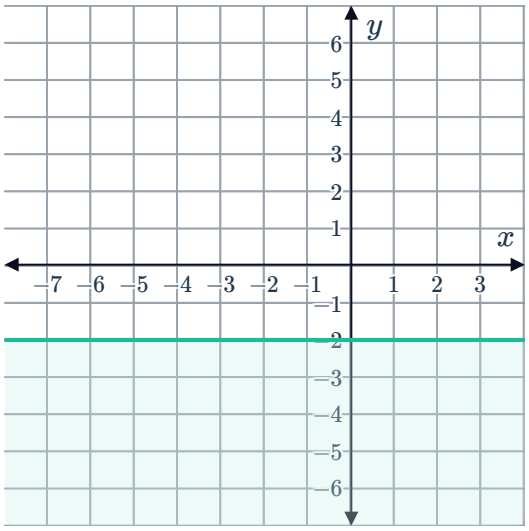


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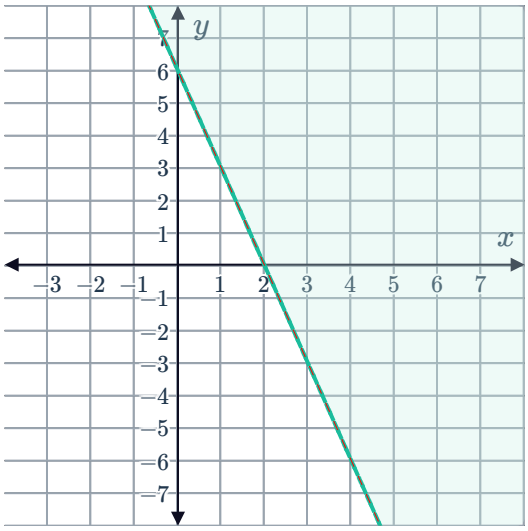


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i



j



4 Consider the equation $2x + 5y = 10$:

- **a** State the x-coordinate of the x-intercept.
- **b** State the y-coordinate of the y-intercept.
- **c** Graph the line on a coordinate plane.
- **d** Write the inequality that describes the region on the graph where $(-20, 11)$ is located.

5 Consider the line $4x + 3y = 12$:

- **a** State the x-coordinate of the x-intercept.
- **b** State the y-coordinate of the y-intercept.
- **c** Graph the solutions to $4x + 3y > 12$.
- **d** Do the points on the line satisfy the inequality $4x + 3y > 12$?

6 Consider the line $y = -2x + 2$:

- **a** State the x-coordinate of the x-intercept.
- **b** State the y-coordinate of the y-intercept.
- **c** State whether the following points satisfies the inequality $y \leq -2x + 2$:

  i $(2, 3)$ ii $(3, -6)$ iii $(4, -2)$ iv $(1, 2)$

- **d** Graph the solutions to $y \leq -2x + 2$.
- **e** Do the points on the line satisfy the inequality $y \leq -2x + 2$?

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7 Consider the line $y = -3x - 6$:

a State the x-coordinate of the x-intercept.

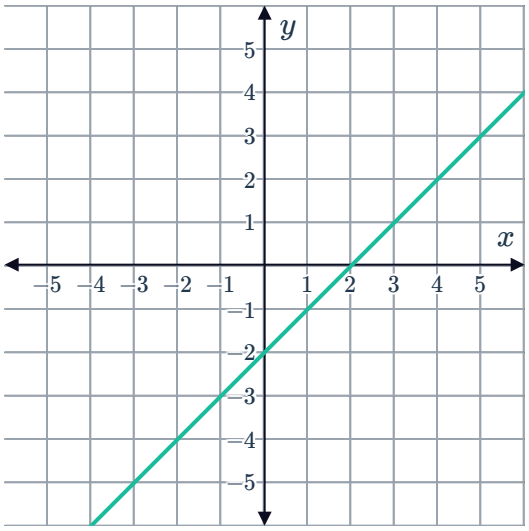
b State the y-coordinate of the y-intercept.

c State whether each of the following points satisfies the inequality $y < -3x - 6$:

i $(-7, 2)$ **ii** $(-6, 3)$ **iii** $(3, -2)$ **iv** $(-6, -3)$

d Graph the solutions to $y < -3x - 6$.

8 Consider the graph of $y = x - 2$:



a Graph the solutions to $y > x - 2$.

b Graph the solutions to $y \leq x - 2$.

9 Consider the inequality $y \geq x + 5$:

a Graph $y = x + 5$.

b State whether the following points satisfies $y \geq x + 5$:

i $(3, 10)$ **ii** $(-3, -2)$

c Graph the solutions to $y \geq x + 5$.

10 Consider the inequality $y > 2x - 4$:

a Graph $y = 2x - 4$.

b State whether the following points satisfies $y > 2x - 4$.

i $(1, -4)$ **ii** $(0, -1)$

c Graph the solutions to $y > 2x - 4$.

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11 Consider the inequality $2x + 3y < 12$:

a Graph $2x + 3y = 12$.

b State whether each of the following points satisfies $2x + 3y < 12$:

i $(2, 4)$ **ii** $(-4, 5)$

c Graph the solutions to $2x + 3y < 12$.

12 Consider the equation $6x + 3y = 12$:

a Complete the following table of values:

x	0	1	2
y			

b State whether each of the following points satisfies the inequality $6x + 3y > 12$:

i $(-2, -2)$ **ii** $(3, 1)$ **iii** $(0, -1)$ **iv** $(-1, 2)$

c Graph the solutions to $6x + 3y > 12$.

13 Graph the following inequalities:

a $y \leq 3x - 4$ **b** $y > 2x - 4$ **c** $y < 3x - 9$ **d** $y \geq -2x + 4$

14 Consider each of the following conditions:

i Find the slope of the boundary line.

ii State the equation of the boundary line in the form $y = mx + b$.

iii State the inequality that satisfies the given conditions.

a A linear inequality has solutions of $(0, -5)$ and $(-1, -9)$ on its boundary line and is not satisifed by $(1, 3)$.

b A linear inequality has solutions of $(0, -6)$ and $(-6, -24)$ on its boundary line and is satisifed by $(-3, -10)$.

15 Rochelle is being careful with her spending so she can later purchase a house. She has allocated no more than $\$855$ each month for both travel expenses and eating out. On average, every time she eats out it costs $\$19$, and every travel journey costs $\$9$.
Write an inequality relating the number of times she eats out, x, and the number of her travel journeys each month, y. Express your final answer with y as the subject.

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16 In an office building, an elevator has a maximum carrying capacity of $12\,750$ pounds. This is based on an average female's weight of 150 pounds and an average male's weight of 170 pounds.
Write an inequality relating the number of women, x, and the number of men, y, who can enter the elevator at one time. Express your final answer with y as the subject.

17 Airline passengers are told to be at the gate no less than 20 minutes before a flight's departure. It takes Rochelle 38 minutes to get to the airport, and y minutes to go through check in and security, depending on how busy the airport is.
Write an inequality relating x and y, where x represents the number of minutes left until Rochelle's flight departs. Express your final answer with y as the subject.

18 Tom has several bottles of Fruita and Greatorade that he wants to take to a picnic, but he only has enough room in his backpack for 12 bottles.

a Write an inequality relating number of bottles of Fruita, x, and the number of bottles of Greatorade, y, that Tom takes to the picnic. Express your final answer with y as the subject.

b Graph the solutions to the inequality.

c State whether Tom can take each of the following combinations of bottles to the picnic:

i 7 bottles of Fruita and 8 bottles of Greatorade

ii 3 bottles of Fruita and 1 bottles of Greatorade

iii 3.8 bottles of Fruita and 3.3 bottles of Greatorade

iv 6 bottles of Fruita and 8 bottles of Greatorade

19 Roxanne has set aside $\$14.00$ in her shopping budget for fruit this week. Currently, oranges are on sale for $\$1.40$ each, while apples are on sale for $\$1.75$ each.

a Write an inequality relating the number of oranges, x, and the number of apples, y, that Roxanne buys. Express your final answer with y as the subject.

b Graph the solutions to the inequality.

c State whether Roxanne can buy each of the following combinations of fruits:

i 6 oranges and 7 apples

ii 3 oranges and 4 apples

iii 7 oranges and 4 apples

iv 1.5 oranges and 2.1 apples

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20 A book seller makes a profit of $\$8$ on every book sold online and $\$5$ on every book sold in store. The company wants to make a profit of at least $\$240$ a day selling books through online and in-store sales.

a Write an inequality relating the number of books sold online, x, the number of books sold in store, y , and the target profit. Express your final answer with y as the subject.

b Graph the solutions to the inequality.

c State whether the following combination of books sold in-store and online will be enough for the book seller to meet the target profit:

i 15.1 online and 29.1 in-store

ii 25 online and 5 in-store

iii 13 online and 23 in-store

iv 20 online and 20 in-store

d Using the graph, find the minimum number of total books (from both online and in store) that need to be sold to reach the target profit.

21 In the new basketball season, William is looking to beat his personal best of scoring 102 points in total for the whole season (this does not include points from fouls).

a If he is to score x 'two pointers' and y 'three pointers' throughout the new season, write an inequality relating x, y and the number of points he hopes to score. Express your final answer with y as the subject.

b Graph the solutions to the inequality.

c In which scenario below will William beat his personal best for the season?

A He scores 17.6 'two pointers' and 27.9 'three pointers'

B He scores 23 'two pointers' and 24 'three pointers'

C He scores 4 'two pointers' and 27 'three pointers'

D He scores 31 'two pointers' and 8 'three pointers'

22 Airline passengers are told to be at the gate no less than 15 minutes before a flight's departure. It takes Gwen 30 minutes to get to the airport, and y minutes to go through check-in and security, depending on how busy the airport is.

a Write an inequality relating x and y, where x represents the number of minutes left until Gwen's flight departs. Express your final answer with y as the subject.

b Graph the solutions to the inequality.

c If the check-in and security process takes 20 minutes, how many minutes before her flight's departure can Gwen leave her house at the latest?

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Additional questions

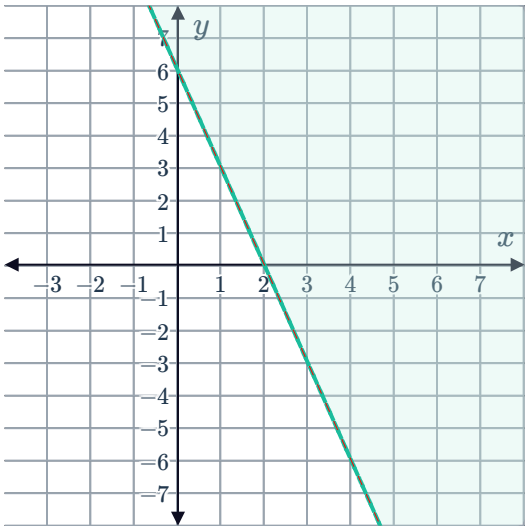
23 For each of the following inequalities, state whether they are represented using a solid or dashed line on a graph.

a $y \leq 4x - 4$ **b** $y < 3x - 5$ **c** $y \geq 2x - 3$ **d** $y > 5$

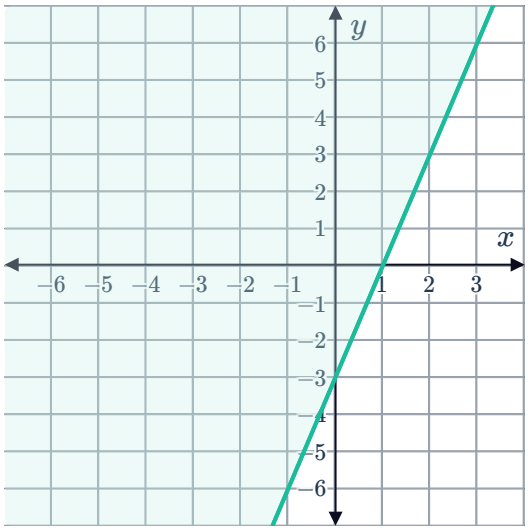
24 State which of the following points lie in each of the given shaded regions:

- $(2, 4)$
- $(-3, 2)$
- $(-1, -6)$
- $(3, -3)$

a



b



25 For each of the following inequalities:

i Determine if the origin $(0, 0)$ satisfies the inequality or not.

ii Sketch the region that satisfies the inequality.

a $x > 3$ **b** $y \leq -2$ **c** $x \geq -1$

26 Applicants for a particular job are asked to sit a numeracy test and verbal reasoning test. Successful applicants must obtain a minimum combined score of 43 for both tests. Write an inequality relating the applicant's score on the numeracy test, x, and the verbal reasoning test, y.

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27 Throughout university, Luigi works as a mentor, getting paid $\$10$ per hour, and as a barista getting paid $\$13$ per hour. The number of hours he works in each job can vary from week to week, and he needs to be able to at least cover his weekly expenses of $\$260$.
Write an inequality relating the number of hours he works as a mentor, x, and barista, y.

28 Oprah is thinking about how to use her last arcade tokens. A game of Table Tennis costs 2 tokens and a game of Frog Days costs 5 tokens. She has a total of 30 tokens left.

a Write an inequality relating the number of times Oprah plays Table Tennis, x and the number of times she plays Frog Days, y.

b Construct a graph of the region containing the points corresponding to all the different ways Oprah could spend her remaining tokens across the two games.

c Interpret the value of the x-intercept of the boundary line in the given context.

d Interpret the value of the y-intercept of the boundary line in the given context.

29 Sean is thinking about how to use his remaining spending money for snacks. A pack of dried fruit costs $\$6$ and a bag of mixed nuts costs $\$4$. He has $\$48$ remaining.

a Write an inequality relating the number of packs of dried fruit, x and the number of bags of mixed nuts, y that Sean buys.

b Sketch the region containing the points corresponding to all the different ways Sean could spend his remaining spending money on these snacks.

c Supposing that Sean does not need to spend all of it, how many different ways can Sean spend his remaining money on some combination of dried fruit and mixed nuts?

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